

The n -Dimensional Burnt Pancake Graph is n^* -connected

Li-Ping Yu, Lih-Hsing Hsu*

Department of Computer Science and Information Engineering
Providence University, Taichung, Taiwan 43301, R.O.C.

Abstract

A k -container $C(u, v)$ of a graph G is a set of k -disjoint paths joining u to v . A k -container $C(u, v)$ of G is k^* -container if $C(u, v)$ spans G . A graph G is k^* -connected if there exists a k^* -container between any two distinct vertices. Let $\kappa(G)$ be the connectivity of G . In this paper, we prove that the n -dimensional burnt pancake graph B_n is n^* -connected for any positive integer n .

1 Introduction

The architecture of an *interconnection network* is usually represented as a *graph* where the vertices represent the processor and the edges represent the links between processors. For the graph definitions and notations, we follow [1]. Let $G = (V, E)$ be a graph if V is a finite set and E is a subset of $\{(a, b) \mid (a, b) \text{ is an unordered pair of } V\}$. We say that V is the *vertex set* and E is the *edge set*. Two vertices u and v are *adjacent* if $(u, v) \in E$. We use $Nbd_G(u)$ to denote the set $\{v \mid (u, v) \in E(G)\}$. The *degree* of a vertex u in G , denoted by $\deg_G(u)$, is $|Nbd_G(u)|$. We use $\delta(G)$ to denote $\min\{\deg_G(u) \mid u \in V(G)\}$. A graph is k -regular if $\deg_G(u) = k$ for every vertex u in G . A path is a sequence of adjacent vertices, written as $\langle v_0, v_1, \dots, v_m \rangle$, in which all the vertices $v_0, v_1, \dots, v_m$ are distinct except that possibly $v_0 = v_m$. We also write the path $\langle v_0, P, v_m \rangle$, where $P = \langle v_0, v_1, \dots, v_m \rangle$. The *length* of a path P , denoted by $l(P)$, is the number of edges in P . Let u and v be two vertices of G . The *distance* between u and v denoted by $d(u, v)$ is the length of the shortest path of G joining u and v . A *cycle* is a path with at least three vertices such that the first vertex is the same as the last one. A *hamiltonian cycle* is a cycle of length $V(G)$. A *hamiltonian path* is a path of length $V(G) - 1$.

Connectivity is an important issue for interconnection networks. The *connectivity* of a graph G , $\kappa(G)$, is the minimum number of vertices whose removal leaves the remaining graph disconnected or trivial. Assume that G is a k -connected graph. It follows from Menger's Theorem that there are k *internally vertex-disjoint* (abbreviated as *disjoint*) paths joining any two distinct vertices u and v [15]. A k -container $C(u, v)$ of G is a set of k disjoint paths joining u to v . Here, we discuss another type of container, called spanning container. A *spanning k -container*, (abbreviated as k^* -container), $C(u, v)$ is a k -container such that it contains all vertices of G . A graph G is k^* -connected if there exists a k^* -container between any two distinct vertices. In particular, a graph G is 1^* -connected if and only if it is hamiltonian connected. Obviously, spanning connectivity is a hybrid concept of hamiltonicity and connectivity. Recently, there are a lot of studies on spanning connectivity [2, 10–14, 16, 17]. The burnt pancake graphs B_n is proposed by Gates and Papadimitriou [4]. After that, many interesting properties of the burnt pancake graphs are studied [3, 7–9]. In particular, the burnt pancake graph can be used for the genome analysis [5]. In this paper, we prove that the n -dimensional burnt pancake graph B_n is n^* -spanning connected for any positive integer n .

2 The burnt pancake graph and its properties

Let n be a positive integer. We use $\langle n \rangle$ to denote the set $\{1, 2, \dots, n\}$. **To save space, the negative sign may be placed on the top of an expression.** Thus, $\bar{u}_1 = -u_1$. We use $[n]$ to denote the set $\langle n \rangle \cup \{\bar{i} \mid i \in \langle n \rangle\}$. A *signed permutation* of $\langle n \rangle$ is an n -permutation $u_1 u_2 \dots u_n$ of $[n]$ such that $|u_1| |u_2| \dots |u_n|$, taking the absolute value of each element, forms a permutation of $\langle n \rangle$. For example, 132654 is a signed permutation of $\langle 6 \rangle$. We will use bold face to denote any

*Corresponding author: lhhsu@cs.pu.edu.tw

signed permutation of $\langle n \rangle$. Hence, $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ denote a sequence of signed permutation of $\langle n \rangle$. Let $\mathbf{u} = u_1 u_2 \dots u_n$ be a signed permutation of $\langle n \rangle$. We use $(\mathbf{u})_i$ to denote the i -th component u_i of $\mathbf{u}$. For $1 \leq i \leq n$, the i -th prefix reversal of $\mathbf{u}$, denoted by $(\mathbf{u})^i$, is the signed permutation $\mathbf{v} = v_1 v_2 \dots v_n$ with $v_j = -u_{i-j+1}$ for $1 \leq j \leq i$ and $v_j = u_j$ if otherwise. For example, $(132654)^4 = 623154$. Thus, $((\mathbf{u})^i)^i = \mathbf{u}$. The n -dimensional burnt pancake graph B_n is a graph containing all the signed permutation of $\langle n \rangle$. Two vertices $\mathbf{u}$ and $\mathbf{v}$ are adjacent in B_n if and only if $\mathbf{v} = (\mathbf{u})^i$. The burnt pancake graph B_1 , B_2 , and B_3 are shown in Figure 1.

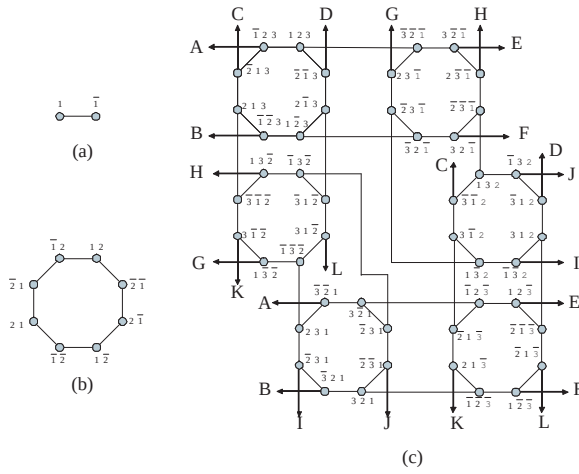


Figure 1: The burnt pancake graphs (a) B_1 , (b) B_2 , and (c) B_3 .

Obviously, B_n is an n -regular graph with $2^n n!$ vertices. We will use $B_n^{\{i\}}$ to denote the i -th subgraph of B_n induced by those vertices $\mathbf{u}$ with $(\mathbf{u})_n = i$. Obviously, B_n can be decomposed into $2n$ vertex disjoint subgraphs $B_n^{\{i\}}$ for every $i \in [n]$ such that each $B_n^{\{i\}}$ is isomorphic to B_{n-1} . Thus, the burnt pancake graph can be constructed recursively. Let $H \subseteq [n]$, we use B_n^H to denote the subgraph of B_n induced by $\cup_{i \in H} V(B_n^{\{i\}})$. For $1 \leq i, j \leq n$ and $i \neq j$, we use $E^{i,j}$ to denote the set of edges between $B_n^{\{i\}}$ and $B_n^{\{j\}}$.

It is easy to check the following Lemma.

Lemma 1. Assume that $n \geq 2$. Then $|E^{i,j}| = 2^{n-2}(n-2)!$ if $1 \leq |i| \neq |j| \leq n$. Moreover, $|E^{i,i}| = 0$ for any i with $1 \leq i \leq n$.

The following Theorem is proved in [9].

Theorem 1. B_n is 1^* -connected if $n \neq 2$, and B_n is 2^* -connected if $n \geq 2$.

Lemma 2. Let $\mathbf{u}$ and $\mathbf{v}$ be any two distinct vertices of B_n with $d(\mathbf{u}, \mathbf{v}) \leq 2$. Then $(\mathbf{u})_1 \neq (\mathbf{v})_1$. Furthermore, $\{ |(\mathbf{u})^i|_1 | 1 \leq i \leq n \} = \langle n \rangle$.

Lemma 3. Let $n \geq 4$ and $i_1, i_2, \dots, i_m$ be an m -permutation of $[n]$ such that $i_k \neq -i_{k+1}$ for $1 \leq k < m$. Then there is a hamiltonian path of $B_n^{\{i_1, i_2, \dots, i_m\}}$ joining any vertex $\mathbf{u} \in V(B_n^{\{i_1\}})$ to any other vertex $\mathbf{v} \in V(B_n^{\{i_m\}})$.

Proof. We set $\mathbf{x}_1 = \mathbf{u}$ and $\mathbf{y}_m = \mathbf{v}$. By Theorem 1, this lemma holds for $m = 1$. Assume that $m \geq 2$. By Lemma 1, we choose $(\mathbf{y}_j, \mathbf{x}_{j+1}) \in E^{i_j, i_{j+1}}$ with $\mathbf{y}_j \neq \mathbf{x}_j$ and $\mathbf{y}_m \neq \mathbf{x}_m$ for every $1 \leq j \leq m-1$. By Theorem 1, there is a hamiltonian path Q_j of $B_n^{\{i_j\}}$ joining $\mathbf{x}_j$ to $\mathbf{y}_j$ for every $1 \leq j \leq m$. The path $\langle \mathbf{x}_1, Q_1, \mathbf{y}_1, \mathbf{x}_2, Q_2, \mathbf{y}_2, \dots, \mathbf{x}_m, Q_m, \mathbf{y}_m \rangle$ forms a desired path.

Let I be a subset of $[n]$. We use $D(I)$ to denote $\{ |j| | j \in \langle n \rangle \text{ such that } \{j, \bar{j}\} \subset I \}$. We have the following lemma.

Lemma 4. Suppose that I be a subset of $[n]$ with $D(I) \geq 2$. Then there exists a hamiltonian path of B_n^I joining any vertex $\mathbf{u} \in V(B_n^{\{i\}})$ to any vertex $\mathbf{v} \in V(B_n^{\{j\}})$ with $\{i, j\} \subset I$ and $|i| \neq |j|$.

Proof. The proof follows from Lemma 3 if we can construct the required permutation of elements in I . We only use several examples to illustrate such permutation exists. Let $I = \{1, 2, \bar{1}, \bar{2}\}$. Suppose that $i = 1$ and $j = 2$. Then the corresponding sequence can be $1, \bar{2}, \bar{1}, 2$. Let $I = \{1, 2, 3, \bar{1}, \bar{2}\}$. Suppose that $i = 1$ and $j = 3$. Then the corresponding sequence can be $1, \bar{2}, \bar{1}, 2, 3$.

Similarly, we have the following lemma.

Lemma 5. Suppose that I is a subset of $[n]$ with $|I| \geq 5$. Then there exists a hamiltonian path of B_n^I joining any vertex $\mathbf{u} \in V(B_n^{\{i\}})$ to any vertex $\mathbf{v} \in V(B_n^{\{j\}})$ with $\{i, j\} \subset I$ and $i \neq j$.

Lemma 6. Let $n \geq 4$. Let $\mathbf{u}$ and $\mathbf{v}$ be any two distinct vertices in $B_n^{\{t\}}$ for some $t \in [n]$. Suppose that B_{n-1} is k^* -connected. Then there is a $(k+1)^*$ -container of B_n between $\mathbf{u}$ and $\mathbf{v}$.

Proof. By assumption, there is a k^* -container $\{Q_1, Q_2, \dots, Q_k\}$ of $B_n^{\{t\}}$ joining $\mathbf{u}$ to $\mathbf{v}$. We need to find a $(k+1)^*$ -container of B_n joining $\mathbf{u}$ to $\mathbf{v}$.

Suppose that $(\mathbf{u})_1 = (\mathbf{v})_1 = p$. Thus, $(\mathbf{u})^n$ and $(\mathbf{v})^n$ are two distinct vertices in $B_n^{\{\bar{p}\}}$. By Lemma 1, there is a hamiltonian path Q of $B_n^{\{\bar{p}\}}$ joining $(\mathbf{u})^n$ to $(\mathbf{v})^n$. We write Q as $\langle (\mathbf{u})^n, Q', \mathbf{y}, \mathbf{z}, (\mathbf{v})^n \rangle$. By Lemma 2, $(\mathbf{y})_1 \neq (\mathbf{z})_1$, $(\mathbf{y})_1 \neq \bar{t}$, and

$(z)_1 \neq \bar{t}$. Since $n \geq 4$, $|[n] - \{t, \bar{p}\}| \geq 6$. By Lemma 5, there exists a hamiltonian path R of $B_n^{[n] - \{t, \bar{p}\}}$ joining $(y)^n$ to $(z)^n$. We set Q_{k+1} as $\langle u, (u)^n, Q', y, (y)^n, R, (z)^n, z, (v)^n, v \rangle$. Obviously, $\{Q_1, Q_2, \dots, Q_{k+1}\}$ forms a $(k+1)^*$ -container of B_n joining u to v . See Figure 2 (a) for illustration.

Suppose that $(u)_1 \neq (v)_1$. Thus, $(u)^n$ and $(v)^n$ are in different subgraphs. Obviously, $|[n] - \{t\}| \geq 7$. By Lemma 5, there is a hamiltonian path Q of $B_n^{[n] - \{t\}}$ joining $(u)^n$ to $(v)^n$. We set Q_{k+1} as $\langle u, (u)^n, Q, (v)^n, v \rangle$. Obviously, $\{Q_1, Q_2, \dots, Q_{k+1}\}$ forms a $(k+1)^*$ -container of B_n joining u to v . See Figure 2 (b) for illustration.

Thus, the lemma is proved.

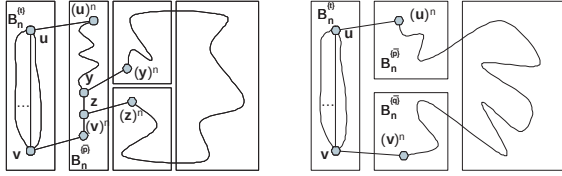


Figure 2: Illustration for Lemma 6.

3 Basic Lemmas

The following lemma is a well-known result that gives a necessary and sufficient condition for a system of distinct representative.

Lemma 7. [6] Let $\mathcal{A} = \{A_1, A_2, \dots, A_m\}$ be a collection of sets. There exists $\{x_1, x_2, \dots, x_m\}$ such that $x_i \in A_i$ for $1 \leq i \leq m$ and $x_i \neq x_j$ if $i \neq j$ if and only if $|\cup_{i \in J} A_i| \geq |J|$ for all $J \subseteq \{1, 2, \dots, m\}$.

Lemma 8. Assume that n and k are positive integers with $n \geq 4$ and $3 \leq k \leq n - 1$. Let s and t are two different elements in $[n]$. Let $x_1, x_2, \dots, x_k$ are k different vertices in B_n with $(x_1)_1 = \bar{s}$, $(x_i)_n \neq \bar{t}$ for $1 \leq i \leq k$, $(x_i)_n \neq (x_j)_n$ for $2 \leq i \leq k$, and $|(x_i)_n| \neq |(x_j)_n|$ for $2 \leq i \neq j \leq k$. Let $y_1, y_2, \dots, y_k$ are k different vertices in B_n with $(y_1)_1 = \bar{t}$, $(y_i)_n \neq \bar{s}$ for $1 \leq i \leq k$, $(y_i)_n \neq (y_j)_n$ for $2 \leq i \leq k$, and $|(y_i)_n| \neq |(y_j)_n|$ for $2 \leq i \neq j \leq k$. Suppose that $\{(x_i)_n \mid 1 \leq i \leq k\} \neq \{(y_i)_n \mid 1 \leq i \leq k\}$. Then there exists k disjoint paths $P_1, P_2, \dots, P_k$ such that (1) P_i joining x_i to $y_{\pi(i)}$ for some permutation π from the set $\{1, 2, \dots, k\}$ into itself and (2) $\cup_{i=1}^k P_i$ spans $B_n^{[n] - \{s, t\}}$.

Proof. Obviously, $x_i \notin V(B_n^{\{s, t\}})$ and $(x_i)^n \in V(B_n^{\{s\}})$. Similarly, $y_i \notin V(B_n^{\{s, t\}})$ and $(y_i)^n \in V(B_n^{\{t\}})$. Let I be the set $\{(x_i)_n \mid (x_i)_n = (y_j)_n \text{ for some } 1 \leq i, j \leq k\}$. We can reorder the indices of $\{1, 2, \dots, k\}$ so that $(x_i)_n = (y_i)_n$ for $1 \leq i \leq |I|$. By Lemma 3, there exists a hamiltonian path P_i of $B_n^{\{(x_i)_n\}}$ joining x_i to y_i for $1 \leq i \leq |I|$. Since $\{(x_i)_n \mid 1 \leq i \leq k\} \neq \{(y_i)_n \mid 1 \leq i \leq k\}$, $|I| < k$.

For $|I| + 1 \leq i \leq k$, let $A_i = \{y_j \mid |I| + 1 \leq j \leq k \text{ with } (x_i)_n \neq (y_j)_n\}$. Obviously, $|A_i| \geq k - 1 - |I|$. Thus, $|\cup_{i \in J} A_i| \geq k - 1 - |I| \geq |J|$ if $\emptyset \neq J \subset \{|I| + 1, |I| + 2, \dots, k\}$. Since $\{(x_i)_n \mid 1 \leq i \leq k\} \neq \{(y_i)_n \mid 1 \leq i \leq k\}$, $|\cup_{i=|I|+1}^k A_i| = k - |I|$. By Lemma 7, there exists $\{y_i \mid |I| + 1 \leq i \leq k\}$ such that $(x_i)_n \neq (y_i)_n$ and $y_i \neq y_j$ for $i \neq j$. Let X be $\{(x_i)_n \mid 1 \leq i \leq k - 1\} \cup \{(y_i)_n \mid 1 \leq i \leq k - 1\} \cup \{s, t\}$.

Suppose that $D([n] - X) \neq 1$. By Lemma 3, there exists a hamiltonian path P_i of $B_n^{\{(x_i)_n, (y_i)_n\}}$ joining x_i to y_i for $|I| + 1 \leq i \leq k - 1$. By Lemma 4 and Lemma 5, there exists a hamiltonian path P_k of $B_n^{[n] - X}$ joining x_k to y_k . Obviously, $\{P_1, P_2, \dots, P_k\}$ forms the required paths.

Suppose that $D([n] - X) = 1$. We claim that $|I| < k - 1$. Suppose not. Then $\{(x_k)_n, -(x_k)_n, (y_k)_n, -(y_k)_n\} \subset ([n] - X)$ and $D([n] - X) \geq 2$. We get a contradiction. Let m be the only positive integer such that $\{m, \bar{m}\} \subset [n] - X$. Then there exists a hamiltonian path P_i of $B_n^{\{(x_i)_n, (y_i)_n\}}$ joining x_i to y_i for $|I| + 1 \leq i \leq k - 2$. Moreover, there exists a hamiltonian path P_{k-1} of $B_n^{\{(x_{k-1})_n, m, (y_{k-1})_n\}}$ joining x_{k-1} to y_{k-1} . Furthermore, there exists a hamiltonian path P_k of $B_n^{([n] - X) - \{m\}}$ joining x_k to y_k . Obviously, $\{P_1, P_2, \dots, P_k\}$ forms the required paths.

Lemma 9. Let $n \geq 4$ and k be any positive integer with $3 \leq k \leq n - 1$. Let u be any vertex in $B_n^{\{s\}}$ and v be any vertex in $B_n^{\{t\}}$ such that $s \neq t$. Suppose that $x_1, x_2, \dots, x_k$ are k vertices in B_n with $(x_1)_1 = \bar{s}$ and $(x_i)_n \neq \bar{t}$ for $1 \leq i \leq k$; and $y_1, y_2, \dots, y_k$ are k vertices in B_n with $(y_1)_1 = \bar{t}$ and $(y_i)_n \neq \bar{s}$ for $1 \leq i \leq k$. Suppose that there exist a permutation π on $\{1, 2, \dots, k\}$ and k disjoint paths, $P_1, P_2, \dots, P_k$, such that P_i is a path joining x_i to $y_{\pi(i)}$ for $1 \leq i \leq k$ and $\cup_{i=1}^k P_i$ spans $B_n^{[n] - \{s, t\}}$. Moreover, there are k internal disjoint paths, $S_1, S_2, \dots, S_k$, of $B_n^{\{s\}}$ such that S_i is a path joining u to $(x_i)^n$ and $\cup_{i=1}^k S_i$ spans $B_n^{\{s\}}$. Furthermore, there are k internal disjoint paths, $T_1, T_2, \dots, T_k$, of $B_n^{\{t\}}$ such that T_i is a path join-

ing $\mathbf{v}$ to $(\mathbf{y}_i)^n$ and $\cup_{i=1}^k T_i$ spans $B_n^{\{t\}}$. Then there exists a k^* -container of B_n joining $\mathbf{u}$ to $\mathbf{v}$. Moreover, this k^* -container does not contain the edge $(\mathbf{u}, \mathbf{v})$ if $(\mathbf{u}, \mathbf{v}) \in E(B_n)$.

Proof. We set Q_i as $\langle \mathbf{u}, S_i, (\mathbf{x}_i)^n, \mathbf{x}_i, P_i, \mathbf{y}_{\pi(i)}, (\mathbf{y}_{\pi(i)})^n, T_{\pi(i)}, \mathbf{v} \rangle$ for $1 \leq i \leq k$. Obviously, $\{Q_1, Q_2, \dots, Q_k\}$ forms the required k^* -container between $\mathbf{u}$ and $\mathbf{v}$.

Lemma 10. Let $n \geq 4$ and k be any positive integer with $3 \leq k \leq n-1$. Let $\mathbf{u}$ be any vertex in $B_n^{\{s\}}$ and $\mathbf{v}$ be any vertex in $B_n^{\{t\}}$ such that $s \neq t$. Suppose that B_{n-1} is k^* -connected. Then there is a k^* -container of B_n between $\mathbf{u}$ and $\mathbf{v}$. Moreover, this k^* -container does not contain the edge $(\mathbf{u}, \mathbf{v})$ if $(\mathbf{u}, \mathbf{v}) \in E(B_n)$.

Proof. Let r be any element in $[n] - \{s, \bar{s}, t, \bar{t}\}$. Suppose that $s \neq \bar{t}$. We set $\mathbf{z}$ to be a vertex with $(\mathbf{z})_1 = r$, $(\mathbf{z})_2 = \bar{t}$, $(\mathbf{z})_n = s$, and $\mathbf{z} \neq \mathbf{u}$; and set $\mathbf{w}$ to be a vertex with $(\mathbf{w})_1 = r$, $(\mathbf{w})_2 = \bar{s}$, $(\mathbf{w})_n = t$, and $\mathbf{w} \neq \mathbf{v}$. Suppose $s = \bar{t}$. We set $\mathbf{z}$ to be a vertex with $(\mathbf{z})_1 = r$, $(\mathbf{z})_n = s$, and $\mathbf{z} \neq \mathbf{u}$; and set $\mathbf{w}$ to be a vertex with $(\mathbf{w})_1 = r$, $(\mathbf{w})_n = t$, and $\mathbf{w} \neq \mathbf{v}$.

Thus, $\mathbf{z} \in V(B_n^{\{s\}})$ and $\mathbf{w} \in V(B_n^{\{t\}})$. By assumption, there exists a k^* -container of $B_n^{\{s\}}$, $\{R_1, R_2, \dots, R_k\}$, joining $\mathbf{u}$ to $\mathbf{z}$. We write $R_i = \langle \mathbf{u}, R'_i, \mathbf{z}_i, \mathbf{z} \rangle$. (Note that $\mathbf{z}_i = \mathbf{u}$ if $l(R_i) = 1$.) Obviously, $((\mathbf{z}_i)_1) \notin \{s, t\}$ for $1 \leq i \leq k$. By Lemma 2, $|(\mathbf{z}_i)_1| \neq |(\mathbf{z}_j)_1|$ for $1 \leq i \neq j \leq k$. Again, there exists a k^* -container of $B_n^{\{t\}}$, $\{H_1, H_2, \dots, H_k\}$, joining $\mathbf{w}$ to $\mathbf{v}$. We write $H_i = \langle \mathbf{w}, \mathbf{w}_i, H'_i, \mathbf{v} \rangle$. (Note that $\mathbf{w}_i = \mathbf{v}$ if $l(H_i) = 1$.) Again, $((\mathbf{w}_i)_1) \notin \{s, t\}$ for $1 \leq i \leq k$, and $|(\mathbf{w}_i)_1| \neq |(\mathbf{w}_j)_1|$ for $1 \leq i \neq j \leq k$. We can reorder the indices of $\{1, 2, \dots, k\}$ so that $\{(\mathbf{z}_i)_1 \mid 2 \leq i \leq k\} \neq \{(\mathbf{w}_i)_1 \mid 2 \leq i \leq k\}$.

Let $\mathbf{x}_1 = (\mathbf{z})^n$ and $\mathbf{x}_i = (\mathbf{z}_i)^n$ for $2 \leq i \leq k$. Similarly, let $\mathbf{y}_1 = (\mathbf{w})^n$ and $\mathbf{y}_i = (\mathbf{w}_i)^n$ for $2 \leq i \leq k$. Obviously, $(\mathbf{x}_i)_1 = \bar{s}$, $(\mathbf{x}_i)_n \neq t$ for $1 \leq i \leq k$, $(\mathbf{x}_i)_n \neq (\mathbf{x}_1)_n$ for $2 \leq i \leq k$, and $|(\mathbf{x}_i)_n| \neq |(\mathbf{x}_j)_n|$ for $2 \leq i \neq j \leq k$. Moreover, $(\mathbf{y}_i)_1 = \bar{t}$, $(\mathbf{y}_i)_n \neq s$ for $1 \leq i \leq k$, $(\mathbf{y}_i)_n \neq (\mathbf{y}_1)_n$ for $2 \leq i \leq k$, and $|(\mathbf{y}_i)_n| \neq |(\mathbf{y}_j)_n|$ for $2 \leq i \neq j \leq k$. Furthermore, $(\mathbf{x}_1)_n = (\mathbf{y}_1)_n = \bar{r}$, and $\{(\mathbf{x}_i)_n \mid 1 \leq i \leq k\} \neq \{(\mathbf{y}_i)_n \mid 1 \leq i \leq k\}$. By Lemma 8, there exist a permutation π and k disjoint paths, $P_1, P_2, \dots, P_k$, such that P_i joining $\mathbf{x}_i$ to $\mathbf{y}_{\pi(i)}$ and $\cup_{i=1}^k P_i$ spans $B_n^{[n]-\{s, t\}}$. By assumption, there exist k internal disjoint paths, $S_1, S_2, \dots, S_k$, such that S_i is a path joining $\mathbf{u}$ to $(\mathbf{x}_i)^n$ for $1 \leq i \leq k$ and $\cup_{i=1}^k S_i$ spans $B_n^{\{s\}}$. Similarly, there exist k internal disjoint paths,

$T_1, T_2, \dots, T_k$, such that T_i is a path joining $(\mathbf{y}_i)^n$ to $\mathbf{v}$ for $1 \leq i \leq k$ and $\cup_{i=1}^k T_i$ spans $B_n^{\{t\}}$. By Lemma 9, the required k^* -container between $\mathbf{u}$ and $\mathbf{v}$ exists. The lemma is proved.

Lemma 11. Assume that $n \geq 4$. Let $\mathbf{u}$ be any vertex in $B_n^{\{s\}}$ with $(\mathbf{u})^n \in V(B_n^{\{t\}})$ and $\mathbf{v}$ be any vertex in $B_n^{\{t\}}$ with $\mathbf{v} \neq (\mathbf{u})^n$ and $(\mathbf{v})^n \in V(B_n^{\{s\}})$. Suppose that $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{n-2}$ are $(n-2)$ vertices in B_n with $(\mathbf{x}_i)_1 = \bar{s}$ and $(\mathbf{x}_i)_n \neq \bar{t}$ for $1 \leq i \leq n-2$; and $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_{n-2}$ are $(n-2)$ vertices in B_n with $(\mathbf{y}_i)_1 = \bar{t}$ and $(\mathbf{y}_i)_n \neq \bar{s}$ for $1 \leq i \leq n-2$. Suppose that there exist a permutation π on $\{1, 2, \dots, n-2\}$ and $(n-2)$ disjoint paths, $P_1, P_2, \dots, P_{n-2}$, such that P_i is a path joining $\mathbf{x}_i$ to $\mathbf{y}_{\pi(i)}$ for $1 \leq i \leq n-2$ and $\cup_{i=1}^{n-2} P_i$ spans $B_n^{[n]-\{s, t\}}$. Moreover, there are $(n-1)$ internal disjoint paths, $S_1, S_2, \dots, S_{n-1}$, of $B_n^{\{s\}}$ such that S_i is a path joining $\mathbf{u}$ to $(\mathbf{x}_i)^n$ for $1 \leq i \leq n-2$, S_{n-1} is a path joining $\mathbf{u}$ to $(\mathbf{v})^n$, and $\cup_{i=1}^{n-1} S_i$ spans $B_n^{\{s\}}$. Furthermore, there are $(n-1)$ internal disjoint paths, $T_1, T_2, \dots, T_{n-1}$, of $B_n^{\{t\}}$ such that T_i is a path joining $(\mathbf{y}_i)^n$ to $\mathbf{v}$ for $1 \leq i \leq n-2$, T_{n-1} is a path joining $(\mathbf{u})^n$ to $\mathbf{v}$, and $\cup_{i=1}^{n-1} T_i$ spans $B_n^{\{t\}}$. Then there exists an n^* -container of B_n joining $\mathbf{u}$ to $\mathbf{v}$.

Proof. Let $Q_i = \langle \mathbf{u}, S_i, (\mathbf{x}_i)^n, \mathbf{x}_i, P_i, \mathbf{y}_{\pi(i)}, (\mathbf{y}_{\pi(i)})^n, T_{\pi(i)}, \mathbf{v} \rangle$ for $1 \leq i \leq n-2$, $Q_{n-1} = \langle \mathbf{u}, (\mathbf{u})^n, T_{n-1}, \mathbf{v} \rangle$, and $Q_n = \langle \mathbf{u}, S_{n-1}, (\mathbf{v})^n, \mathbf{v} \rangle$. Then $\{Q_1, Q_2, \dots, Q_n\}$ forms an n^* -container between $\mathbf{u}$ and $\mathbf{v}$. The lemma is proved.

Lemma 12. Assume that $n \geq 4$. Let $\mathbf{u}$ be any vertex in $B_n^{\{s\}}$ with $(\mathbf{u})^n \in V(B_n^{\{t\}})$ and $\mathbf{v}$ be any vertex in $B_n^{\{t\}}$ with $\mathbf{v} \neq (\mathbf{u})^n$ and $(\mathbf{v})^n \in V(B_n^{\{s\}})$. Suppose that B_{n-1} is $(n-1)^*$ -connected. Then there is an n^* -container of B_n between $\mathbf{u}$ and $\mathbf{v}$.

Proof. Obviously, $|s| \neq |t|$. By assumption, there exists an $(n-1)^*$ -container of $B_n^{\{s\}}$, $\{R_1, R_2, \dots, R_{n-1}\}$, joining $\mathbf{u}$ to $(\mathbf{v})^n$. We write $R_i = \langle \mathbf{u}, R'_i, \mathbf{z}_i, (\mathbf{v})^n \rangle$. (Note that $\mathbf{z}_i = \mathbf{u}$ if $l(R_i) = 1$.) Obviously, $(\mathbf{z}_i)_1 \notin \{s, t\}$ for $1 \leq i \leq n-1$. By Lemma 2, $|(\mathbf{z}_i)_1| \neq |(\mathbf{z}_j)_1|$ for $1 \leq i \neq j \leq n-1$. Again, there exists an $(n-1)^*$ -container of $B_n^{\{t\}}$, $\{H_1, H_2, \dots, H_{n-1}\}$, joining $(\mathbf{u})^n$ to $\mathbf{v}$. We write $H_i = \langle (\mathbf{u})^n, \mathbf{w}_i, H'_i, \mathbf{v} \rangle$. (Note that $\mathbf{w}_i = \mathbf{v}$ if $l(H_i) = 1$.) Again, $(\mathbf{w}_i)_1 \notin \{s, t\}$ for $1 \leq i \leq n-1$, and $|(\mathbf{w}_i)_1| \neq |(\mathbf{w}_j)_1|$ for $1 \leq i \neq j \leq n-1$. We can reorder the indices of $\{1, 2, \dots, n-1\}$ so that $\{(\mathbf{z}_i)_1 \mid 1 \leq i \leq n-2\} \neq \{(\mathbf{w}_i)_1 \mid 1 \leq i \leq n-2\}$.

Let $\mathbf{x}_i = (\mathbf{z}_i)^n$ for $1 \leq i \leq n-2$ and $\mathbf{y}_i = (\mathbf{w}_i)^n$ for $1 \leq i \leq n-2$. Obviously, $(\mathbf{x}_i)_1 = s$, $(\mathbf{x}_i)_n \neq t$ for $1 \leq i \leq n-2$, $|(\mathbf{x}_i)_n| \neq |(\mathbf{x}_j)_n|$ for $1 \leq i \neq j \leq n-2$. Moreover, $(\mathbf{y}_i)_1 = t$, $(\mathbf{y}_i)_n \neq s$ for $1 \leq i \leq n-2$, and $|(\mathbf{y}_i)_n| \neq |(\mathbf{y}_j)_n|$ for $1 \leq i \neq j \leq n-2$. Furthermore, $\{(\mathbf{x}_i)_n \mid 1 \leq i \leq n-2\} \neq \{(\mathbf{y}_i)_n \mid 1 \leq i \leq n-2\}$. By Lemma 8, there exist a permutation π on $\{1, 2, \dots, n-2\}$ and $(n-2)$ disjoint paths, $P_1, P_2, \dots, P_{n-2}$, such that P_i joining $\mathbf{x}_i$ to $\mathbf{y}_{\pi(i)}$ and $\cup_{i=1}^{n-2} P_i$ spans $B_n^{[n]-\{s,t\}}$. By assumption, there exist $(n-1)$ internal disjoint paths, $S_1, S_2, \dots, S_{n-1}$, such that S_i is a path joining $\mathbf{u}$ to $(\mathbf{x}_i)^n$ for $1 \leq i \leq n-1$ and $\cup_{i=1}^{n-1} S_i$ spans $B_n^{\{s\}}$. Similarly, there exist $(n-1)$ internal disjoint paths, $T_1, T_2, \dots, T_{n-1}$, such that T_i is a path joining $(\mathbf{y}_i)^n$ to $\mathbf{v}$ for $1 \leq i \leq n-1$ and $\cup_{i=1}^{n-1} T_i$ spans $B_n^{\{t\}}$. By Lemma 11, there exists an n^* -container between $\mathbf{u}$ and $\mathbf{v}$. The lemma is proved.

Lemma 13. Assume that $n \geq 4$. Let $\mathbf{u}$ be any vertex in $B_n^{\{s\}}$ with $(\mathbf{u})^n \in V(B_n^{\{t\}})$ and $\mathbf{v}$ be any vertex in $B_n^{\{t\}}$ with $\mathbf{v} \neq (\mathbf{u})^n$ and $(\mathbf{v})^n \notin V(B_n^{\{s\}})$. Suppose that $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{n-1}$ are $(n-1)$ vertices in B_n with $(\mathbf{x}_i)_1 = \bar{s}$ and $(\mathbf{x}_i)_n \neq \bar{t}$ for $1 \leq i \leq n-1$; and $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_{n-1}$ are $(n-1)$ vertices in B_n with $(\mathbf{y}_i)_1 = \bar{t}$ and $(\mathbf{y}_i)_n \neq \bar{s}$ for $1 \leq i \leq n-1$. Moreover, $(\mathbf{v})^n = \mathbf{y}_{n-1}$. Suppose that there exist a permutation π on $\{1, 2, \dots, n-1\}$ and $(n-1)$ disjoint paths, $P_1, P_2, \dots, P_{n-1}$, such that P_i is a path joining $\mathbf{x}_i$ to $\mathbf{y}_{\pi(i)}$ for $1 \leq i \leq n-1$ and $\cup_{i=1}^{n-1} P_i$ spans $B_n^{[n]-\{s,t\}}$. Moreover, there are $(n-1)$ internal disjoint paths, $S_1, S_2, \dots, S_{n-1}$, of $B_n^{\{s\}}$ such that S_i is a path joining $\mathbf{u}$ to $(\mathbf{x}_i)^n$ for $1 \leq i \leq n-1$ and $\cup_{i=1}^{n-1} S_i$ spans $B_n^{\{s\}}$. Furthermore, there are $(n-1)$ internal disjoint paths, $T_1, T_2, \dots, T_{n-1}$, of $B_n^{\{t\}}$ such that T_i is a path joining $\mathbf{v}$ to $(\mathbf{y}_i)^n$ for $1 \leq i \leq n-2$, T_{n-1} is a path joining $(\mathbf{u})^n$ to $\mathbf{v}$, and $\cup_{i=1}^{n-1} T_i$ spans $B_n^{\{t\}}$. Then there exists an n^* -container of B_n joining $\mathbf{u}$ to $\mathbf{v}$.

Proof. Without loss of generality, we assume that $\pi(n-1) = n-1$. Let $Q_i = \langle \mathbf{u}, S_i, (\mathbf{x}_i)^n, \mathbf{x}_i, P_i, \mathbf{y}_{\pi(i)}, (\mathbf{y}_{\pi(i)})^n, T_{\pi(i)}, \mathbf{v} \rangle$ for $1 \leq i \leq n-2$, $Q_{n-1} = \langle \mathbf{u}, S_{n-1}, (\mathbf{x}_{n-1})^n, \mathbf{x}_{n-1}, P_{n-1}, \mathbf{y}_{n-1} = (\mathbf{v})^n, \mathbf{v} \rangle$, and $Q_n = \langle \mathbf{u}, (\mathbf{u})^n, T_{n-1}, \mathbf{v} \rangle$. Then $\{Q_1, Q_2, \dots, Q_n\}$ forms an n^* -container between $\mathbf{u}$ and $\mathbf{v}$. The lemma is proved.

Lemma 14. Assume that $n \geq 4$. Let $\mathbf{u}$ be any vertex in $B_n^{\{s\}}$ with $(\mathbf{u})^n \in V(B_n^{\{t\}})$ and $\mathbf{v}$ be any vertex in $B_n^{\{t\}}$ with $\mathbf{v} \neq (\mathbf{u})^n$ and $(\mathbf{v})^n \notin V(B_n^{\{s\}})$. Suppose that B_{n-1} is $(n-1)^*$ -connected. Then there is an n^* -container of B_n between $\mathbf{u}$ and $\mathbf{v}$.

Proof. Since $(\mathbf{u})^n \in V(B_n^{\{t\}})$, $|s| \neq |t|$ and $(\mathbf{u})_1 = \bar{t}$. We set $\mathbf{z}$ be the vertex with $(\mathbf{z})_1 = t$ and $(\mathbf{z})_i = (\mathbf{u})_i$ for $2 \leq i \leq n$. Thus, $\mathbf{z} \in V(B_n^{\{s\}})$.

By assumption, there exists an $(n-1)^*$ -container of $B_n^{\{s\}}$, $\{R_1, R_2, \dots, R_{n-1}\}$, joining $\mathbf{u}$ to $\mathbf{z}$. We write $R_i = \langle \mathbf{u}, R'_i, \mathbf{z}_i, \mathbf{z} \rangle$. (Note that $\mathbf{z}_i = \mathbf{u}$ if $l(R_i) = 1$.) Obviously, $(\mathbf{z}_i)_1 \notin \{s, t\}$ for $1 \leq i \leq n-1$. By Lemma 2, $|(\mathbf{z}_i)_1| \neq |(\mathbf{z}_j)_1|$ for $1 \leq i \neq j \leq n-1$. Again, there exists an $(n-1)^*$ -container of $B_n^{\{t\}}$, $\{H_1, H_2, \dots, H_{n-1}\}$, joining $(\mathbf{u})^n$ to $\mathbf{v}$. We write $H_i = \langle (\mathbf{u})^n, \mathbf{w}_i, H'_i, \mathbf{v} \rangle$. (Note that $\mathbf{w}_i = \mathbf{v}$ if $l(H_i) = 1$.) Again, $(\mathbf{w}_i)_1 \notin \{s, t\}$ for $1 \leq i \leq n-1$, and $|(\mathbf{w}_i)_1| \neq |(\mathbf{w}_j)_1|$ for $1 \leq i \neq j \leq n-1$. We can reorder the indices of $\{1, 2, \dots, n-1\}$ so that $\{(\mathbf{z}_i)_1 \mid 2 \leq i \leq n-1\} \neq \{(\mathbf{w}_i)_1 \mid 2 \leq i \leq n-1\}$.

Let $\mathbf{x}_1 = (\mathbf{z})^n$ and $\mathbf{x}_i = (\mathbf{z}_i)^n$ for $2 \leq i \leq n-1$. Similarly, $\mathbf{y}_1 = (\mathbf{v})^n$ and $\mathbf{y}_i = (\mathbf{w}_i)^n$ for $2 \leq i \leq n-1$. Obviously, $(\mathbf{x}_i)_1 = s$, $(\mathbf{x}_i)_n \neq t$ for $1 \leq i \leq n-1$, $(\mathbf{x}_i)_n \neq (\mathbf{x}_j)_n$ for $2 \leq i \neq j \leq n-1$, and $|(\mathbf{x}_i)_n| \neq |(\mathbf{x}_j)_n|$ for $2 \leq i \neq j \leq n-1$. Moreover, $(\mathbf{y}_i)_1 = t$, $(\mathbf{y}_i)_n \neq s$ for $1 \leq i \leq n-1$, $(\mathbf{y}_i)_n \neq (\mathbf{y}_j)_n$ for $2 \leq i \neq j \leq n-1$, and $|(\mathbf{y}_i)_n| \neq |(\mathbf{y}_j)_n|$ for $2 \leq i \neq j \leq n-1$. Furthermore, $(\mathbf{x}_1)_n = (\mathbf{y}_1)_n = \bar{t}$, and $\{(\mathbf{x}_i)_n \mid 1 \leq i \leq n-1\} \neq \{(\mathbf{y}_i)_n \mid 1 \leq i \leq n-1\}$.

By Lemma 8, there exist a permutation π on $\{1, 2, \dots, n-1\}$ and $(n-1)$ disjoint paths, $P_1, P_2, \dots, P_{n-1}$, such that P_i joining $\mathbf{x}_i$ to $\mathbf{y}_{\pi(i)}$ and $\cup_{i=1}^{n-1} P_i$ spans $B_n^{[n]-\{s,t\}}$. By assumption, there exist $(n-1)$ internal disjoint paths, $S_1, S_2, \dots, S_{n-1}$, such that S_i is a path joining $\mathbf{u}$ to $(\mathbf{x}_i)^n$ for $1 \leq i \leq n-1$ and $\cup_{i=1}^{n-1} S_i$ spans $B_n^{\{s\}}$. Similarly, there exist $(n-1)$ internal disjoint paths, $T_1, T_2, \dots, T_{n-1}$, such that T_i is a path joining $(\mathbf{y}_i)^n$ to $\mathbf{v}$ for $1 \leq i \leq n-1$ and $\cup_{i=1}^{n-1} T_i$ spans $B_n^{\{t\}}$. By Lemma 13, there exists an n^* -container between $\mathbf{u}$ and $\mathbf{v}$. The lemma is proved.

Similar to Lemma 13, we have the following lemma.

Lemma 15. Assume that $n \geq 4$. Let $\mathbf{u}$ be a vertex in $B_n^{\{s\}}$ and $\mathbf{v}$ be a vertex in $B_n^{\{t\}}$ with $s \neq t$, $(\mathbf{u})^n \notin V(B_n^{\{t\}})$, and $(\mathbf{v})^n \notin V(B_n^{\{s\}})$. Suppose that $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ are n vertices in B_n with $(\mathbf{x}_i)_1 = \bar{s}$, $(\mathbf{x}_i)_n \neq \bar{t}$ for $1 \leq i \leq n$; and $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n$ are n vertices in B_n with $(\mathbf{y}_i)_1 = \bar{t}$, $(\mathbf{y}_i)_n \neq \bar{s}$ for $1 \leq i \leq n$. Moreover, $(\mathbf{u})^n = \mathbf{x}_n$ and $(\mathbf{v})^n = \mathbf{y}_n$. Suppose that there exist a permutation on $\{1, 2, \dots, n\}$ and n disjoint paths, $P_1, P_2, \dots, P_n$, such that P_i is a path joining $\mathbf{x}_i$ to $\mathbf{y}_{\pi(i)}$ for $1 \leq i \leq n$ and $\cup_{i=1}^n P_i$ spans $B_n^{[n]-\{s,t\}}$. Moreover, there are $(n-1)$ internal disjoint paths,

$S_1, S_2, \dots, S_{n-1}$, of $B_n^{\{s\}}$ such that S_i is a path joining $\mathbf{u}$ to $(\mathbf{x}_i)^n$ for $1 \leq i \leq n-1$ and $\cup_{i=1}^{n-1} S_i$ spans $B_n^{\{s\}}$. Furthermore, there are $(n-1)$ internal disjoint paths, $T_1, T_2, \dots, T_{n-1}$, of $B_n^{\{t\}}$ such that T_i is a path joining $\mathbf{v}$ to $(\mathbf{y}_i)^n$ for $1 \leq i \leq n-1$, and $\cup_{i=1}^{n-1} T_i$ spans $B_n^{\{t\}}$. Then there exists an n^* -container of B_n joining $\mathbf{u}$ to $\mathbf{v}$.

Lemma 16. Assume that n is a positive integer with $n \geq 4$. Let s and t are two different elements in $[n]$. Let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ are n different vertices in B_n with $(\mathbf{x}_i)_1 = \bar{s}$, $(\mathbf{x}_i)_n \neq \bar{t}$ for $1 \leq i \leq n$, and $(\mathbf{x}_i)_n \neq (\mathbf{x}_j)_n$ for $1 \leq i \neq j \leq n$. Let $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n$ are n different vertices in B_n with $(\mathbf{y}_i)_1 = \bar{t}$, $(\mathbf{y}_i)_n \neq \bar{s}$ for $1 \leq i \leq n$, and $(\mathbf{y}_i)_n \neq (\mathbf{y}_j)_n$ for $1 \leq i \neq j \leq n$. Suppose that $\{(\mathbf{x}_i)_n \mid 1 \leq i \leq n\} \neq \{(\mathbf{y}_i)_n \mid 1 \leq i \leq n\}$. Moreover, $\{ |(\mathbf{x}_i)_n| \mid 1 \leq i \leq n \} = \langle n \rangle - \{|s|\}$ and $\{ |(\mathbf{y}_i)_n| \mid 1 \leq i \leq n \} = \langle n \rangle - \{|t|\}$. Then there exist a permutation π on $\{1, 2, \dots, n\}$ and n disjoint paths $P_1, P_2, \dots, P_n$ such that P_i joining $\mathbf{x}_i$ to $\mathbf{y}_{\pi(i)}$ for $1 \leq i \leq n$ and $\cup_{i=1}^n P_i$ spans $B_n^{[n] - \{s, t\}}$.

Proof. Obviously, $\mathbf{x}_i \notin V(B_n^{\{s, t\}})$ and $(\mathbf{x}_i)^n \in V(B_n^{\{s\}})$. Similarly, $\mathbf{y}_i \notin V(B_n^{\{s, t\}})$ and $(\mathbf{y}_i)^n \in V(B_n^{\{t\}})$. Let I be the set $\{(\mathbf{x}_i)_n \mid (\mathbf{x}_i)_n = (\mathbf{y}_j)_n \text{ for some } 1 \leq i, j \leq k\}$. We can reorder the indices of $\{1, 2, \dots, n\}$ so that $(\mathbf{x}_i)_n = (\mathbf{y}_i)_n$ for $1 \leq i \leq |I|$. By Theorem 1, there exists a hamiltonian path P_i of $B_n^{\{(\mathbf{x}_i)_n\}}$ joining $\mathbf{x}_i$ to $\mathbf{y}_i$ for $1 \leq i \leq |I|$.

For $|I|+1 \leq i \leq n$, let $A_i = \{\mathbf{y}_j \mid |I|+1 \leq j \leq n \text{ with } (\mathbf{x}_i)_n \neq (\mathbf{y}_j)_n\}$. By Lemma 7, there exists $\{\mathbf{y}_i \mid |I|+1 \leq i \leq n\}$ such that $(\mathbf{x}_i)_n \neq (\mathbf{y}_i)_n$ and $\mathbf{y}_i \neq \mathbf{y}_j$ for $i \neq j$. Let X be $\{(\mathbf{x}_i)_n \mid 1 \leq i \leq n-1\} \cup \{(\mathbf{y}_i)_n \mid 1 \leq i \leq n-1\} \cup \{s, t\}$. Moreover, $\{ |(\mathbf{x}_i)_n| \mid 1 \leq i \leq n \} = \langle n \rangle - \{|s|\}$ and $\{ |(\mathbf{y}_i)_n| \mid 1 \leq i \leq n \} = \langle n \rangle - \{|t|\}$. Obviously, $D([n] - X) = 0$. By Lemma 3, there exists a hamiltonian path P_i of $B_n^{\{(\mathbf{x}_i)_n, (\mathbf{y}_i)_n\}}$ joining $\mathbf{x}_i$ to $\mathbf{y}_i$ for $|I|+1 \leq i \leq n-1$. Again, there exists a hamiltonian path P_n of $B_n^{[n] - X}$ joining $\mathbf{x}_n$ to $\mathbf{y}_n$. Obviously, $\{P_1, P_2, \dots, P_n\}$ forms the required paths.

Lemma 17. Assume that $n \geq 4$. Let $\mathbf{u}$ be a vertex in $B_n^{\{s\}}$ and $\mathbf{v}$ be a vertex in $B_n^{\{t\}}$ with $s \neq t$, $(\mathbf{u})^n \notin V(B_n^{\{t\}})$, and $(\mathbf{v})^n \notin V(B_n^{\{s\}})$. Suppose that B_{n-1} is $(n-1)^*$ -connected. Then there is an n^* -container of B_n between $\mathbf{u}$ and $\mathbf{v}$.

Proof. Suppose that $s \neq \bar{t}$. We set $\mathbf{z}$ be a vertex with $(\mathbf{z})_1 = t$, $(\mathbf{z})_2 = (\mathbf{u})_1$, and $(\mathbf{z})_n = s$; and set $\mathbf{w}$ be a vertex with $(\mathbf{w})_1 = s$, $(\mathbf{w})_2 = (\mathbf{v})_1$, and $(\mathbf{w})_n = t$. Suppose $s = \bar{t}$. Since $n \geq 4$, there exists an element r in $\langle n \rangle - \{|s|, |(\mathbf{u})_1|, |(\mathbf{v})_1|\}$. We

set $\mathbf{z}$ be a vertex with $(\mathbf{z})_1 = r$, $(\mathbf{z})_2 = (\mathbf{u})_1$, and $(\mathbf{z})_n = s$; and set $\mathbf{w}$ be a vertex with $(\mathbf{w})_1 = \bar{r}$, $(\mathbf{w})_2 = (\mathbf{v})_1$, and $(\mathbf{w})_n = t$.

By assumption, there exists an $(n-1)^*$ -container of $B_n^{\{s\}}$, $\{R_1, R_2, \dots, R_{n-1}\}$, joining $\mathbf{u}$ to $\mathbf{z}$. We write $R_i = \langle \mathbf{u}, R'_i, \mathbf{z}, \mathbf{z} \rangle$. (Note that $\mathbf{z}_i = \mathbf{u}$ if $l(R_i) = 1$.) Obviously, $(\mathbf{z}_i)_1 \notin \{s, t\}$ for $1 \leq i \leq n-1$. By Lemma 2, $|(\mathbf{z}_i)_1| \neq |(\mathbf{z}_j)_1|$ for $1 \leq i \neq j \leq n-1$. Again, there exists an $(n-1)^*$ -container of $B_n^{\{t\}}$, $\{H_1, H_2, \dots, H_{n-1}\}$, joining $\mathbf{w}$ to $\mathbf{v}$. We write $H_i = \langle \mathbf{w}, \mathbf{w}_i, H'_i, \mathbf{v} \rangle$. (Note that $\mathbf{w}_i = \mathbf{v}$ if $l(H_i) = 1$.) Again, $(\mathbf{w}_i)_1 \notin \{s, t\}$ for $1 \leq i \leq n-1$, and $|(\mathbf{w}_i)_1| \neq |(\mathbf{w}_j)_1|$ for $1 \leq i \neq j \leq n-1$. We can reorder the indices of $\{1, 2, \dots, n-1\}$ so that $\{ |(\mathbf{z}_i)_1| \mid 2 \leq i \leq n-1 \} = \{ |(\mathbf{w}_i)_1| \mid 2 \leq i \leq n-1 \} = \langle n \rangle - \{|s|, |t|\}$ if $s \neq \bar{t}$; and $\{ |(\mathbf{z}_i)_1| \mid 2 \leq i \leq n-1 \} = \{ |(\mathbf{w}_i)_1| \mid 2 \leq i \leq n-1 \} = \langle n \rangle - \{|s|, |r|\}$ if otherwise.

Let $\mathbf{x}_1 = (\mathbf{z})^n$, $\mathbf{x}_i = (\mathbf{z}_i)^n$ for $2 \leq i \leq n-1$ and $\mathbf{x}_n = (\mathbf{u})^n$. Similarly, $\mathbf{y}_1 = (\mathbf{w})^n$, $\mathbf{y}_i = (\mathbf{w}_i)^n$ for $2 \leq i \leq n-1$ and $\mathbf{y}_n = (\mathbf{v})^n$. Obviously, $(\mathbf{x}_i)_1 = s$, $(\mathbf{x}_i)_n \neq t$ for $1 \leq i \leq n$, $(\mathbf{x}_i)_n \neq (\mathbf{x}_j)_n$ for $2 \leq i \leq n$, and $|(\mathbf{x}_i)_n| \neq |(\mathbf{x}_j)_n|$ for $2 \leq i \neq j \leq n$. Moreover, $(\mathbf{y}_i)_1 = t$, $(\mathbf{y}_i)_n \neq s$ for $1 \leq i \leq n$, $(\mathbf{y}_i)_n \neq (\mathbf{y}_j)_n$ for $2 \leq i \leq n-1$, and $|(\mathbf{y}_i)_n| \neq |(\mathbf{y}_j)_n|$ for $2 \leq i \neq j \leq n-1$. Note that $(\mathbf{x}_1)_n = \bar{t}$ and $(\mathbf{y}_1)_n = \bar{s}$ if $s \neq t$; and $(\mathbf{x}_1)_n = \bar{r}$ and $(\mathbf{y}_1)_n = r$ if $s = t$. Thus, $\{(\mathbf{x}_i)_n \mid 1 \leq i \leq n\} \neq \{(\mathbf{y}_i)_n \mid 1 \leq i \leq n\}$. Moreover, $\{ |(\mathbf{x}_i)_n| \mid 1 \leq i \leq n \} = \langle n \rangle - \{|s|\}$ and $\{ |(\mathbf{y}_i)_n| \mid 1 \leq i \leq n \} = \langle n \rangle - \{|t|\}$. By Lemma 16, there exist a permutation π on $\{1, 2, \dots, n\}$ and n disjoint paths, $P_1, P_2, \dots, P_n$, such that P_i joining $\mathbf{x}_i$ to $\mathbf{y}_{\pi(i)}$ and $\cup_{i=1}^n P_i$ spans $B_n^{[n] - \{s, t\}}$. By assumption, there exist $(n-1)$ internal disjoint paths, $S_1, S_2, \dots, S_{n-1}$, such that S_i is a path joining $\mathbf{u}$ to $(\mathbf{x}_i)^n$ for $1 \leq i \leq n-1$ and $\cup_{i=1}^{n-1} S_i$ spans $B_n^{\{s\}}$. Similarly, there exist $(n-1)$ internal disjoint paths, $T_1, T_2, \dots, T_{n-1}$, such that T_i is a path joining $(\mathbf{y}_i)^n$ to $\mathbf{v}$ for $1 \leq i \leq n-1$ and $\cup_{i=1}^{n-1} T_i$ spans $B_n^{\{t\}}$. By Lemma 15, there exists an n^* -container between $\mathbf{u}$ and $\mathbf{v}$. The lemma is proved.

4 Main result

Theorem 2. B_n is n^* -connected for any positive integer n .

Proof. We prove this theorem by induction. It is easy to see that B_1 is 1^* -connected and B_2 is 2^* -connected. Since the B_3 is vertex transitive, by brute force we can check that B_3 is 3^* -connected.

Assume that B_k is k^* -connected for every $3 \leq k \leq n-1$. Let $\mathbf{u}$ and $\mathbf{v}$ be any two distinct vertices of B_n with $\mathbf{u} \in V(B_n^{\{s\}})$ and $\mathbf{v} \in V(B_n^{\{t\}})$. We need to find an n^* -container between $\mathbf{u}$ and $\mathbf{v}$ of B_n . Suppose that $s = t$. By induction, there exists an $(n-1)^*$ -container of $B_n^{\{s\}}$ joining $\mathbf{u}$ to $\mathbf{v}$. By Lemma 6, there is an n^* -container of B_n joining $\mathbf{u}$ to $\mathbf{v}$. Thus, we assume that $s \neq t$.

Case 1: $(\mathbf{u})^n \in V(B_n^{\{t\}})$ and $(\mathbf{v})^n \in V(B_n^{\{s\}})$.

Suppose that $\mathbf{u} = (\mathbf{v})^n$. By Lemma 10, there is an $(n-1)^*$ -container $\{Q_1, Q_2, \dots, Q_{n-1}\}$ of B_n joining $\mathbf{u}$ to $\mathbf{v}$ not using the edge $(\mathbf{u}, \mathbf{v})$. We set Q_n as $\langle \mathbf{u}, \mathbf{v} \rangle$. Then $\{Q_1, Q_2, \dots, Q_n\}$ forms an n^* -container of B_n joining $\mathbf{u}$ to $\mathbf{v}$. Suppose that $\mathbf{u} \neq (\mathbf{v})^n$. By Lemma 12, there is an n^* -container of B_n joining $\mathbf{u}$ to $\mathbf{v}$.

Case 2: $((\mathbf{u})^n \in V(B_n^{\{t\}})$ and $(\mathbf{v})^n \notin V(B_n^{\{s\}}))$ or $((\mathbf{u})^n \notin V(B_n^{\{t\}})$ and $(\mathbf{v})^n \in V(B_n^{\{s\}}))$. Without loss of generality, we assume that $(\mathbf{u})^n \in V(B_n^{\{t\}})$ and $(\mathbf{v})^n \notin V(B_n^{\{s\}})$. By Lemma 14, there is an n^* -container of B_n joining $\mathbf{u}$ to $\mathbf{v}$.

Case 3: $(\mathbf{u})^n \notin V(B_n^{\{t\}})$ and $(\mathbf{v})^n \notin V(B_n^{\{s\}})$. By Lemma 17, there is an n^* -container of B_n joining $\mathbf{u}$ to $\mathbf{v}$.

The theorem is proved.

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